

Imprint Field Dynamics: Coherence-Entanglement Continuum and the Reinterpretation of the Double-Slit Experiment

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Keywords: quantum field, coherence, entanglement, double-slit, triadic topology, imprint field, post-quantum

PACS: 03.65.-w, 03.67.-a, 42.50.Pq

Abstract

The Imprint Field Hypothesis reinterprets the quantum measurement process as a coupling between particle

1. Introduction

The double-slit experiment demonstrates the interplay between particle discreteness and wave interference.

2. The Imprint Field Hypothesis

Reality in this model is represented as $R = P + F + I$, where Particle (P), Field (F), and Imprint (I) form the triad.

3. Coherence-Entanglement Continuum (CEC)

CEC describes the smooth transition between coherence and entanglement. The coherence parameter $\gamma = \frac{F}{P}$.

4. Reinterpretation of the Double-Slit Experiment

Under the Imprint Field model, a particle generates a coherent imprint through both slits. The interference pattern is a result of the field's interaction with the imprint.

5. Experimental and Technological Applications

- Quantum Computing: Triadic qutrits with coherence-entanglement binary logic.
- Quantum Sensing: Controlled imprint coupling enhances weak-field sensitivity.
- Quantum Materials: Field-structured lattices maintain coherence, similar to hydrodynamic walking droplets.

6. Discussion and Outlook

The Imprint Field Dynamics reframes quantum behavior as emergent field coherence rather than probabilistic outcomes.

7. Conclusion

Reality is triadic—an interaction between particle, field, and imprint. Controlling the coherence-entanglement transition offers new insights into quantum phenomena.

Appendix: Supplementary Methods

Triadic Field Equations:

$$dP/dt = -i\omega_P P - gPF^*, \quad dF/dt = -i\omega_F F + gPI, \quad dI/dt = -\Gamma I + gF.$$

Steady-state coherence condition ($dI/dt = 0$): $F = (\Gamma/g)I$.

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Submitted to arXiv (Quantum Physics)